REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

Generated: 2026-09-21 14:48:44
Reporting Period: All Recorded Traffic

1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	15587
Normal Flows	14502
Detected Attacks	1085
Attack Rate	6.96%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
HTTPS	5081
TCP	4054
UDP	3482
DNS	1563
QUIC_OR_UDP443	1247
HTTP	78
IPSEC	64
DHCP	14
OTHER	2
ICMP	1
NTP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Threat Vector	Severity	Confidence
2026-09-21 14:43:30	10.118.118.202:51584	35.186.224.35:443	HTTPS	ANOMALOUS FLOW	HIGH	85.0%
2026-09-21 14:43:18	72.153.5.140:443	10.118.118.202:54788	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-21 14:43:18	23.201.53.161:443	10.118.118.202:63768	HTTPS	ANOMALOUS FLOW	HIGH	54.52%
2026-09-21 14:43:17	10.118.118.202:52651	10.2.0.1:65432	TCP	ANOMALOUS FLOW	HIGH	85.0%
2026-09-21 14:43:15	172.67.135.197:443	10.118.118.202:56698	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	73.0%
2026-09-21 14:43:13	10.118.118.202:55726	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:10	10.118.118.202:5353	224.0.0.251:5353	UDP	ANOMALOUS FLOW	HIGH	80.0%
2026-09-21 14:43:08	104.21.7.52:443	10.118.118.202:63915	HTTPS	ANOMALOUS FLOW	HIGH	58.94%
2026-09-21 14:43:08	10.118.118.202:51002	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:07	10.118.118.82:53	10.118.118.202:61052	DNS	ANOMALOUS FLOW	HIGH	92.25%
2026-09-21 14:43:07	10.118.118.82:53	10.118.118.202:54068	DNS	ANOMALOUS FLOW	HIGH	87.55%
2026-09-21 14:43:07	10.118.118.82:53	10.118.118.202:64559	DNS	ANOMALOUS FLOW	HIGH	97.66%
2026-09-21 14:43:06	10.118.118.202:51001	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:61872	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:58883	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:63429	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:57045	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:51496	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:62897	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-21 14:43:06	10.118.118.202:53783	10.118.118.82:53	DNS	ANOMALOUS FLOW	HIGH	95.0%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.